

that plans on bringing down the size of a phase change cooler to that of regular AIO liquid coolers.

Liquid submersion

If you've had the privilege of attending PC component expos or conferences, you must have come across crazy PC builds that don't make sense at all. One of them must have been systems completely submerged under water where the CPU is bubbling. However, that isn't water but a special liquid called Novec by 3M. It's called immersion cooling which is actually targeted towards data centers. The liquid is non-conductive so there aren't any sparks flying around even though it's directly in contact with the electrical components. It's also a good conductor of heat so as it boils it turns into vapour and takes away the heat. They have a low boiling point of around 49 degrees Celsius. After it turns to vapour, it reaches a condenser attached on the top to turn back into liquid and fall down into the same system. Hence, it forms a self-contained



The der8auer Aqua Exhalare case cooled with Novec liquid

cooling system. You need to ensure that your container is leak-proof and houses the condenser on the top. PC cases built for immersion cooling are modded from existing cases to be made commercially available.

Other than the super expensive Novec liquid, you can use another liquid to submerge your system and that is oil. Preferably, mineral oil. It's non-conductive and offers a high rate of heat transfer. However, there are certain things you need to keep in mind if you're using mineral oil. You shouldn't use any kind of rubber materials in any of the components that will be submerged since it tends to degrade under mineral oil. Based on the material of the case, you need to keep a watch on the temperature of the oil. Otherwise, the container might crack with the heat. Oil also expands when it gets heated,